

State Capacity as Adjustment Architecture

Common Shocks, Interoperable Payments, and Social Protection

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Abstract

Governments must allocate administrative investment across shared delivery platforms, local execution capacity, and information systems. Using raw Andhra Pradesh smartcard microdata from the OpenICPSR archive, I re-estimate large NREGS delivery gains: collection time falls by 19.4percent, payment delay by 25.2percent, payment unpredictability by 34.1percent, and work access rises by 16.8percent. The paper’s empirical centerpiece is a reconstructed GP-week panel showing that rollout gains are substantially larger in high-common weeks: the probability of any weekly disbursement or work rises by about 5.6percentage points in low-common weeks and by 15.1percentage points in high-common weeks, while total disbursed amounts and days worked rise more as well. A conditional-on-positive-activity decomposition shows that the overall payout-per-day amplification mainly reflects this extensive-margin activation rather than a clean within-active-week throughput gain. Event-time and placebo evidence are consistent with the recovered timing design, and the strictest same-setting differential remains 0.409 (0.095) after restricting the outcome sample to the 496 GPs originally assigned to treatment, defining high-common weeks only from untreated GPs outside the focal district, and absorbing district-specific month and year effects. SSP and a public correlational state-month panel provide supporting breadth checks rather than separate hard causal pillars. A parsimonious policy model then translates these empirical facts into an administrative portfolio comparison for common-shock environments.

Keywords: state capacity, social protection, digital payments, interoperability, public administration.

JEL: O41, D61, H54, C61.

1 Introduction

A government preparing for the next recession, drought, or disaster faces a specific policy choice before the shock arrives. Should scarce administrative funds go into a shared payment-and-authentication backbone, into more local delivery staff and offices, or into data systems that identify whom to reach once need is realized? Those investments can look similar in the budget today, but they imply very different margins of maneuverability tomorrow. The question is related to, but distinct from, the broader literature on state capacity that studies the determinants and consequences of administrative

capability at a more aggregated level (Besley and Persson, 2009, 2011; Acemoglu et al., 2015; Aneja and Xu, 2024).

The paper’s empirical center is a direct test of the architecture prediction inside Andhra’s smartcard rollout. In a matched ± 26 week window around recovered conversion dates, rollout raises the probability of any weekly disbursement or work by about 5.6percentage points in low-common weeks and by about 15.1percentage points in high-common weeks. The overall GP-week payout-per-day outcome also rises more in high-common weeks, but the same matched panel shows that the amplification primarily comes from fewer zero-activity weeks and larger total disbursed amounts and days worked, not from a clean within-active-week payout-per-day gain. That is a direct test of the paper’s main comparative static inside Andhra data rather than a structural extrapolation: interoperable architecture delivers larger gains when workload shocks are more common. Event-time patterns and placebo conversion dates are consistent with the recovered timing design; the headline differential remains 0.450 (0.088) when the sample is restricted to the 496 GPs originally assigned to treatment, remains 0.519 (0.095) when the commonality measure is built only from not-yet-converted GPs outside the focal district, remains 0.596 (0.097) when both restrictions are imposed simultaneously, remains 0.409 (0.095) after adding district-specific month and year effects to that strictest specification, remains 0.403 with a mandal-clustered standard error of 0.110, and stays between 0.250 and 0.480 when districts are omitted one at a time.

The same repo also re-estimates the underlying Andhra treatment effects from the raw NREGS smartcards files. Collection time falls by 19.4percent, payment delay by 25.2percent, payment unpredictability by 34.1percent, and work access rises by 16.8percent. SSP provides a supporting breadth check in a second program with the same platform intervention but a different beneficiary and payment environment. District-month MGNREGA data identify the covariance environment itself, and a public state-month panel provides the strongest additional supporting breadth check currently feasible with in-repo data. The linked beneficiary records are useful for bounding the information block, but they do not amount to direct registry-to-payment linkage.

Three results emerge. First, the Andhra GP-week data support the core comparative static: interoperable architecture delivers larger gains when workload shocks are common. Second, the raw microdata show policy-relevant delivery improvements in NREGS, and SSP shows that the same platform improved delivery in a distinct transfer program. Third, once those facts are placed inside the Andhra covariance environment, composition mistakes are quantitatively consequential: holding the administrative envelope fixed, importing a more local architecture costs about 2.3percent of permanent consumption, and that ranking survives a wide local-capacity bound search that lets the local-response target rise to 100.0percent.

The model is used only to translate that empirical ranking into an ex ante portfolio choice over interoperable platforms, local execution, and information assets. The common-versus-local ranking remains robust to aggressive local-capacity alternatives, while the information block remains explicitly secondary because the public data do not identify it as cleanly as the GP-week delivery design.

The remainder of the article is organized as follows. Section 2 describes the institutional policy choice in targeted transfers. Section 3 states only the minimum framework needed for the empirical

exercise, with formal derivations and proofs left to the appendix. Section 4 centers the empirical evidence. Section 5 then translates those facts into structural welfare and budgeted architecture comparisons. Section 6 reports robustness and scope conditions. Section 7 concludes.

2 Institutional setting and policy choice

Targeted transfers are a useful setting for studying administrative architecture because the same fiscal objective can be implemented through very different institutional bundles. A government can route transfers through a national payment-and-identity backbone, through more decentralized local delivery staff and offices, or through registries that improve targeting before delivery begins. Those choices are not equivalent. Shared platforms are especially valuable when needs move together. Local delivery is especially valuable when needs diverge across places. Information systems are valuable only insofar as the state can act on what they reveal. These distinctions are central in the design of social protection systems in low- and middle-income countries (Banerjee et al., 2024; Haseeb and Vyborny, 2022).

The Indian transfer setting makes those distinctions concrete. MGNREGA demand is highly seasonal, which means that one can observe whether need is primarily common or locally offsetting. The Andhra smartcard rollout then provides unusually clear evidence on what a shared delivery platform buys in practice: faster payment collection, shorter delays, and lower leakage (Muralidharan et al., 2016). These are precisely the operational margins that the model interprets as actuation capacity. Targeted transfers are therefore not treated as an idiosyncratic special case, but as an application in which the key administrative objects are observable enough to support welfare analysis.

The empirical material combines public administrative data with public replication-package microdata. The district-month panel identifies workload covariance. Raw NREGS and SSP microdata identify the shared platform’s delivery effects. The district interaction design provides a conservative within-setting discipline on local capacity. Linked beneficiary smartcard-status records provide the within-setting information moment. This design identifies the common-platform margin and bounds the local margin conservatively while keeping the remaining information gap visible.

3 Minimal policy framework (formal details in the appendix)

This section keeps only the framework needed for the empirical paper. The appendix contains the general vector-control environment, the full recourse derivation, the information block, and the decentralization problem. In the main text, the relevant policy question is narrower: when should a government invest in a shared interoperable backbone rather than in more local redundancy?

3.1 A minimal two-mode architecture problem

The planner allocates resources across productive capital k_t , common platform capacity g_{Ct} , and local execution capacity g_{Lt} . Output is produced by a standard neoclassical technology $F(k_t)$, and

the slow resource constraint is

$$\dot{k}_t = F(k_t) - c_t - i_{Ct} - i_{Lt} - \delta_k k_t. \quad (3.1)$$

$$\dot{g}_{jt} = i_{jt} - \delta_j g_{jt}, \quad j \in \{C, L\}. \quad (3.2)$$

At Poisson rate λ , a contingency arrives that requires reallocating administrative effort across places. The appendix shows how installed administrative architecture determines the value of fast recourse after that shock is revealed.

To isolate the paper's core comparative static, consider two regions. Let

$$u_+ := \frac{1}{\sqrt{2}}(1, 1)^\top, \quad u_- := \frac{1}{\sqrt{2}}(1, -1)^\top$$

denote the common and differential directions. Suppose desired reallocations have covariance

$$\Sigma = \sigma^2 \left[(1 + \varrho) u_+ u_+^\top + (1 - \varrho) u_- u_-^\top \right], \quad \varrho \in [-1, 1]. \quad (3.3)$$

The parameter ϱ is the correlation of the two regions' desired reallocations. Positive ϱ means that common shocks dominate; negative ϱ means that needs tend to move in opposite directions. A common interoperable platform g_C loads on the common mode with effectiveness $\omega_C > 0$, and local redundant capacity g_L loads on the differential mode with effectiveness $\omega_L > 0$:

$$W(g_C, g_L) = \omega_C g_C u_+ u_+^\top + \omega_L g_L u_- u_-^\top. \quad (3.4)$$

More generally, an installed portfolio of actuation assets generates

$$W(g) := \sum_{j=1}^J g_j W_j. \quad (3.5)$$

The appendix shows that the ex ante recourse value per contingency is

$$\mathcal{R}(g) = \frac{1}{2} \text{tr} \left[\Sigma W(g) (W(g) + \kappa I)^{-1} \right]. \quad (3.6)$$

The next proposition is the only theorem-level comparative static the empirical section needs.

Proposition 3.1 (Common platforms versus local redundancy). *Under (3.3)–(3.4),*

$$\mathcal{R}(g_C, g_L) = \frac{\sigma^2}{2} \left[(1 + \varrho) \frac{\omega_C g_C}{\kappa + \omega_C g_C} + (1 - \varrho) \frac{\omega_L g_L}{\kappa + \omega_L g_L} \right]. \quad (3.7)$$

Therefore

$$\partial_{g_C} \mathcal{R}(g_C, g_L) = \frac{\sigma^2 \kappa}{2} (1 + \varrho) \frac{\omega_C}{(\kappa + \omega_C g_C)^2}, \quad \partial_{g_L} \mathcal{R}(g_C, g_L) = \frac{\sigma^2 \kappa}{2} (1 - \varrho) \frac{\omega_L}{(\kappa + \omega_L g_L)^2}.$$

The marginal value of common capacity is strictly increasing in ϱ , while the marginal value of local redundant capacity is strictly decreasing in ϱ .

At the origin, common capacity is more valuable than local redundant capacity if and only if

$$\varrho > \varrho^* := \frac{\omega_L - \omega_C}{\omega_L + \omega_C}. \quad (3.8)$$

If $\delta_C = \delta_L$ and both assets are active in steady state, then

$$\frac{\kappa + \omega_C g_C^*}{\kappa + \omega_L g_L^*} = \sqrt{\frac{(1 + \varrho)\omega_C}{(1 - \varrho)\omega_L}}, \quad (3.9)$$

so the optimal portfolio tilts toward common platforms as shocks become more correlated.

The proposition gives the paper’s policy core. When shocks are predominantly common, the planner wants more interoperable backbone capacity; when shocks are predominantly idiosyncratic, the planner wants more local redundancy. The empirical job is therefore to learn three objects: how common shocks are, how strongly the shared platform changes delivery, and how much local capacity absorbs idiosyncratic strain.

Information and decentralization. The appendix adds an information stock m that shrinks posterior uncertainty before recourse through

$$\Pi(m) := \left(\Sigma^{-1} + \sum_{\ell=1}^L m_{\ell} K_{\ell} \right)^{-1}, \quad K_{\ell} \succeq 0, \quad (3.10)$$

and shows that its value rises with the delivery architecture it complements. The same appendix also shows why decentralized jurisdictions underprovide the common platform relative to the planner: they do not internalize the full systemwide dividend from the shared backbone. Those results motivate the linked-beneficiary information evidence and the conservative federalism counterfactual later in the paper.

4 Empirical architecture evidence

This section centers the paper’s main empirical claim. It first tests the common-shock comparative static inside the Andhra rollout data, then reports the baseline NREGS delivery gains, and finally shows the strongest additional breadth evidence currently feasible with in-repo data: SSP and a public state-month interaction outside Andhra. The local-capacity and information blocks used in the policy model are deferred to the next section.

4.1 Centerpiece result: rollout gains are larger in high-common weeks

The reconstructed GP-week rollout panel permits a direct empirical test of the common-shock comparative static. In a matched ± 26 week window around recovered GP conversion dates, rollout raises the probability of any weekly disbursement or work by about 5.6percentage points in low-common weeks and by about 15.1percentage points in high-common weeks, using 27010GP-weeks across 560GPs in 8 districts and 53 mandals. 1 shows the normalized event-time patterns around

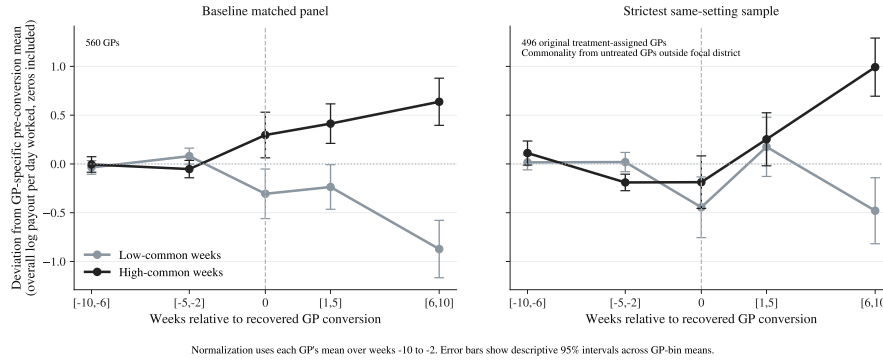


Figure 1. Normalized event-time plots around recovered GP conversion dates in the baseline and strictest Andhra GP-week samples

those recovered conversion dates in both the baseline matched panel and the strictest same-setting sample, and 1 reports the baseline decomposition on any positive payout/work week, total disbursed amounts, days worked, the overall payout-per-day outcome, and the payout-per-day outcome conditional on a positive-work week, followed by the continuous, rank-based, alternative leave-one-out, treated-assignment-only, untreated-only, untreated-outside-district, combined treated-assignment plus untreated-outside-district, strictest same-setting plus district-specific seasonality, geography-and-inference, balanced-window, calendar-window, and placebo checks for the headline overall payout-per-day outcome.

This is the paper’s central empirical result. It tests the model’s common-shock prediction inside the public Andhra rollout files rather than only through the structural mapping. The baseline decomposition in 1 shows that high-common amplification runs through more active payout/work weeks and larger total disbursements and days worked, with the amount response larger than the days-worked response; the descriptive conditional row then shows that the overall payout-per-day amplification does not appear to be a clean within-active-week price effect. 1 complements the table by showing that the post-conversion high-common versus low-common separation is visible not only in the baseline matched panel but also in the stricter treated-assignment sample that defines commonality only from untreated GPs outside the focal district. Restricting the sample to the 496 GPs originally assigned to treatment still yields a differential of 0.450 (0.088), which makes the centerpiece less reliant on later recovered conversions outside the original treated rollout arm. A stricter untreated-only robustness redefines common-workload using only not-yet-converted GPs in the same calendar week, drops weeks with no untreated comparison pool, and still yields a differential of 0.358 (0.086). Building the common-workload measure only from not-yet-converted GPs outside the focal district still yields 0.519 (0.095), which makes the result harder to explain as either treated statewide activity or district-specific workload spikes masquerading as a common shock. Combining both restrictions at once still yields 0.596 (0.097) in 22113 GP-weeks from the 496 original treatment-assigned GPs. Adding district-specific month and year effects to that same strictest specification still yields 0.409 (0.095), which makes the same-setting result harder to explain as a district-specific seasonal rollout pattern layered on top of the staggered timing design. Excluding every GP in the focal district from the common-workload measure still yields 0.509 (0.103). Re-clustering the baseline specification at the mandal level yields 0.403 (0.110), and leaving out

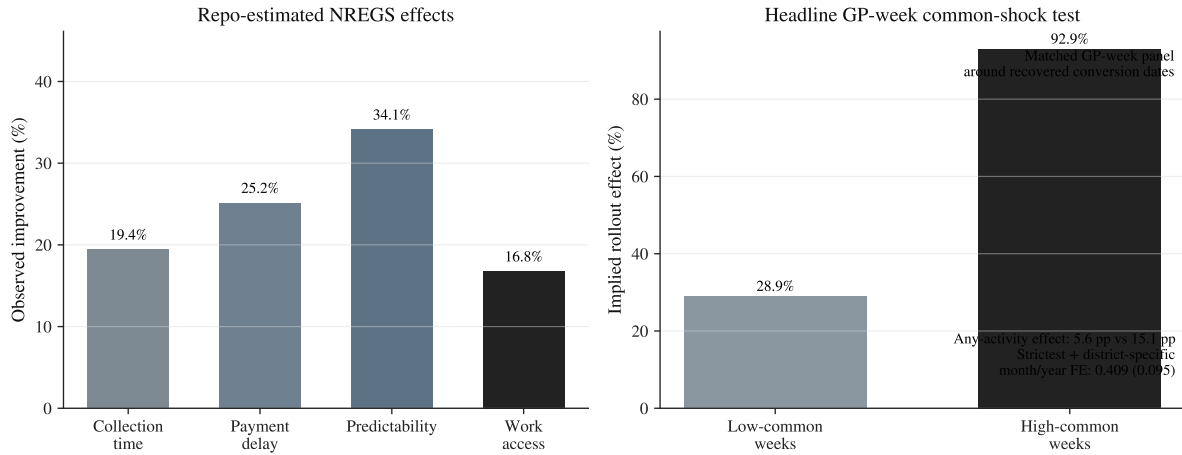


Figure 2. Average Andhra delivery gains and the headline GP-week common-shock test

each district in turn still leaves interactions between 0.250 and 0.480. The balanced-window row requires at least twenty observed pre-conversion and twenty observed post-conversion weeks per GP and yields a similar interaction. The placebo rows in 1 assign pseudo conversion dates twelve and sixteen weeks before actual conversion and yield no comparable interaction, which is consistent with the recovered timing design. A separate timing-balance panel in 11 also shows that, within the original treatment-assigned arm, recovered conversion dates are not meaningfully predicted by baseline collection time, baseline survey payout levels, or baseline work access.

4.2 Direct delivery gains in NREGS

The GP-week result matters because the underlying platform changed first-order delivery outcomes in the raw Andhra microdata. The raw OpenICPSR NREGS files show that interoperable payment and authentication architecture improved observable delivery margins in one application where administrative implementation mattered directly for welfare. 2 summarizes the repo-estimated treatment effects, and 2 puts those average NREGS gains next to the GP-week amplification that is the paper’s main mechanism result.

These margins admit a direct policy interpretation. Faster collection, shorter delays, and more predictable payouts are evidence that a shared platform changed execution quality. Higher received amounts and higher work access show that the result is not limited to process outcomes.

4.3 Breadth beyond the original NREGS design

The strongest additional empirical evidence currently feasible in-repo is breadth rather than a second causal design. First, the same smartcard platform also changed delivery in SSP, which provides a supporting breadth check in a second program context with a different beneficiary and payment environment. 3 shows that official take-up was large in SSP, collection time fell, survey-reported amounts rose, and leakage fell.

Second, a public state-month panel places Andhra inside a broader distribution of common-shock

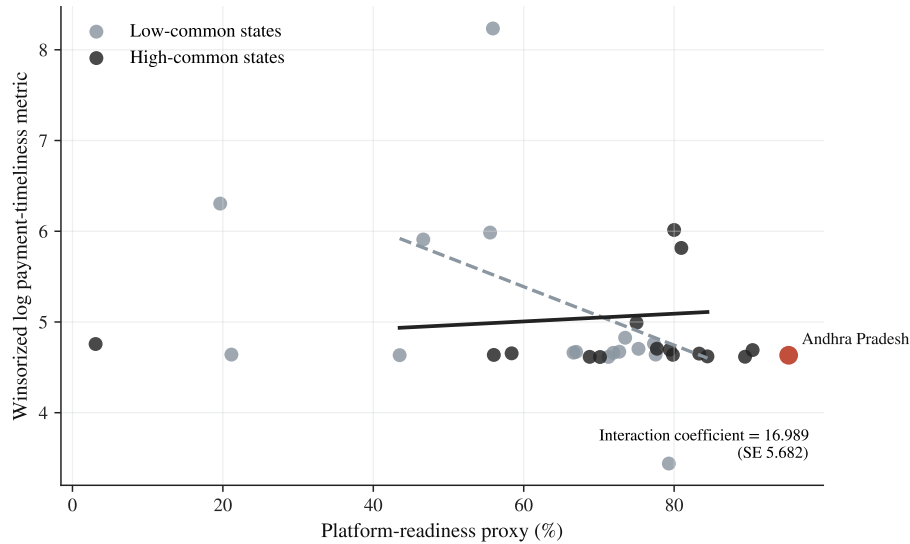


Figure 3. Correlational breadth check: platform readiness is more strongly associated with payment performance in high-common states

and platform-readiness environments and asks whether the same architecture logic appears outside the original NREGS design.

The state-month panel is not a second causal design. It is instead a supporting breadth check showing that platform readiness is more strongly associated with the public payment-timeliness measure in states where workload shocks are more common. Table 4 and Figure 3 report a positive interaction coefficient of 16.989 with a clustered standard error of 5.682. The pattern is correlational, but it shows that the Andhra mechanism aligns with broader state-month relationships rather than only with Andhra’s position in a cross-sectional scatter.

With current repo contents, the SSP second-program result and this state-month panel are the strongest supporting breadth checks available beyond the original NREGS microdata.

5 Policy translation and quantitative comparison

The remaining task is not to establish whether the platform mattered. It is to translate the empirical ranking into a budget-neutral architecture comparison while keeping the weaker objects explicit and conservative. This section therefore uses the strongest same-setting inputs available and treats the local-capacity and information blocks as supporting disciplines rather than as headline designs.

5.1 Supporting linked beneficiary evidence and the information block

The linked beneficiary panel provides the paper’s best feasible in-repo evidence on the information margin. 5 compares smartcard holders with non-holders inside treated carded GPs. Access gains are stronger among smartcard-holder households, and leakage reductions are concentrated there as well. The benchmark therefore uses the smartcard-holder work-access gain as a conservative same-setting information moment and keeps the linked leakage estimates as supporting evidence. This is still not

a transaction-level registry-to-payment linkage design, so the information block remains secondary to the GP-week and average-delivery evidence.

5.2 Objects taken to the model

The empirical exercise estimates the four objects required by the welfare exercise inside one Andhra setting: shock covariance, shared-platform response, local-delivery response, and information/authentication gains. 6 summarizes that mapping.

To keep the benchmark transparent, the quantitative exercise uses a two-region reduction with one common actuation asset, one local actuation asset, and one information asset. Write the common and differential shock variances as

$$\sigma_+^2 = s_\xi^2(1 + \varrho), \quad \sigma_-^2 = s_\xi^2(1 - \varrho), \quad (5.1)$$

and let the information stock contribute mode-specific precision

$$\pi_+(m) = (\sigma_+^{-2} + \eta_+ m)^{-1}, \quad \pi_-(m) = (\sigma_-^{-2} + \eta_- m)^{-1}. \quad (5.2)$$

The benchmark adaptation value is then

$$\mathcal{R}(g_C, g_L, m; \varrho) = \frac{1}{2} \left[(\sigma_+^2 - \pi_+(m)) \frac{\omega_C g_C}{\kappa + \omega_C g_C} + (\sigma_-^2 - \pi_-(m)) \frac{\omega_L g_L}{\kappa + \omega_L g_L} \right]. \quad (5.3)$$

The empirical section uses a permanent-consumption-equivalent welfare metric to compare steady-state architectures. Because productive capital is pinned down by the standard Ramsey steady-state condition, these counterfactuals differ through steady-state consumption and the recourse dividend. The appendix reports both the exact formula and its local approximation; for the equal-budget comparisons emphasized here, the welfare comparison is driven entirely by recourse differences.

5.3 Same-setting Andhra data and identification

The core data object is a district-month Andhra MGNREGA panel built from the public data.gov.in district feed. The panel reports cumulative persondays, households worked, wages, expenditure, and payment-timeliness measures by district, month, and fiscal year. I difference the cumulative series into monthly flows, keep complete fiscal years, and remove district, month, and fiscal-year effects before estimating the covariance of realized workload.

The same public feed also supports the local-capacity design. I construct a pre-period district capacity index from early-sample payment timeliness, work-completion ratios, days of employment per household, and active-worker intensity. The local block is then disciplined by the interaction between idiosyncratic district workload shocks and that pre-period capacity index. Intuitively, higher-capacity districts should absorb locally high workload months with less deterioration in administrative performance if local capacity is a relevant adjustment asset. Because those interaction estimates remain weak, the quantitative section uses them as a benchmark floor plus a wide bound search rather than as a clean positive-identification local design. A separate auxiliary regression

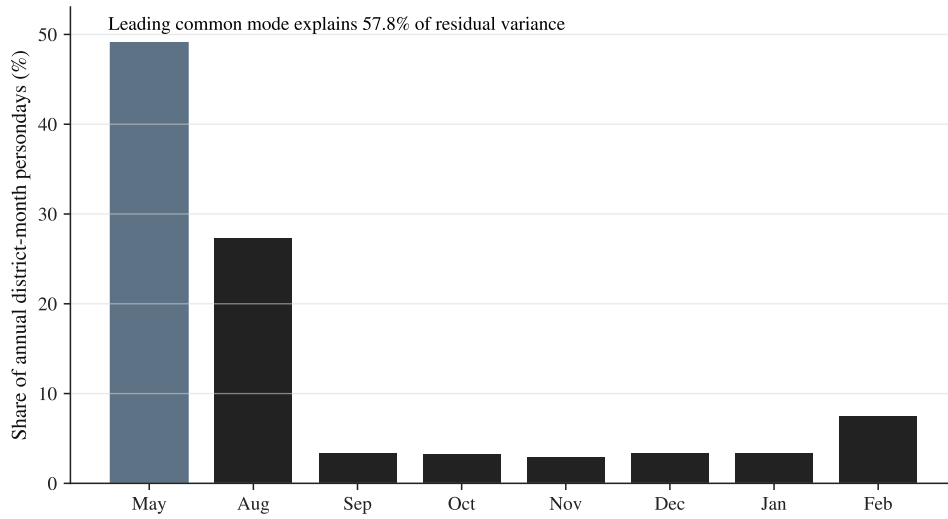


Figure 4. Andhra district-month workload data imply a strongly common shock component

with baseline banking access is reported only as a scope check and does not supply an exogenous local-capacity design.

The shared-platform and information/authentication blocks come from raw Andhra microdata in the OpenICPSR archive. The common-platform object is estimated from the repo’s NREGS treatment effects on collection time, payment delay, and payment predictability. The information object is estimated from linked beneficiary smartcard-status records, using the work-access gain among smartcard-holder households in treated carded GPs as the conservative structural moment and the linked leakage estimates as supporting evidence. That mapping interprets biometric authentication and beneficiary identification as information capital that becomes valuable only when delivery systems can act on it.

5.4 Estimated shock covariance

The district-month panel implies a strongly common-shock Andhra environment. 4 shows the average monthly pattern in personday flows, while the full district covariance matrix implies that the leading common mode explains about 57.8percent of residual workload variance. In the two-mode reduction, that maps into

$$\hat{c}_+ = 57.8/100, \quad \hat{\rho} = 0.561. \quad (5.4)$$

The estimate is stronger than a seasonal concentration summary because it uses the full district-month covariance matrix rather than a single state-level statistic. The economic conclusion is unchanged: Andhra is a high-common-shock environment, not a low-correlation one.

5.5 Delivery and information inputs for the structural mapping

The repo-estimated NREGS effects discipline the common-platform block. In treated NREGS areas, carded payments rose by about 44.7percentage points, while collection time, payment delay, and payment unpredictability all fell materially. I summarize those margins with the within-setting

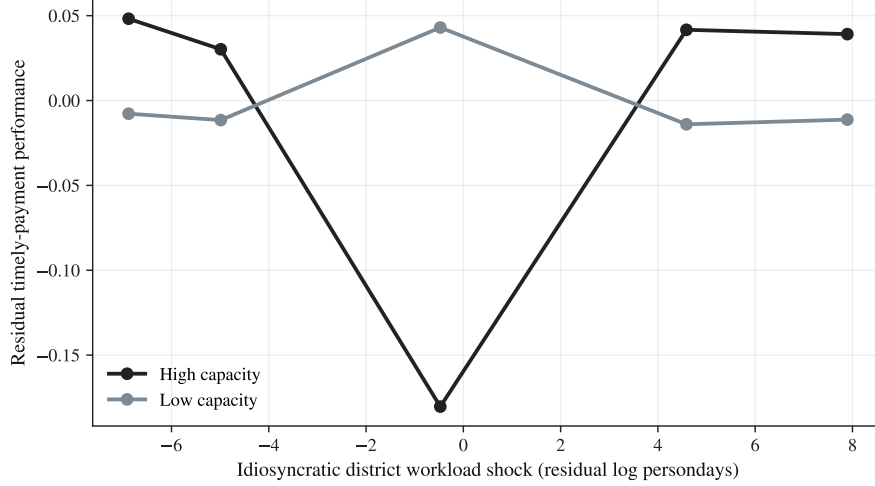


Figure 5. Higher-capacity Andhra districts absorb local workload shocks more effectively

common-platform response index

$$\hat{\psi}_C = 26.2/100, \quad \phi_C = 44.7/100, \quad (5.5)$$

and choose ω_C so that the model-implied effective platform response,

$$\psi_C(\theta) = \phi_C \frac{\omega_C g_C^*(\theta)}{\kappa + \omega_C g_C^*(\theta)}, \quad (5.6)$$

matches that observed operational improvement.

The local block is disciplined by a within-setting interaction design. Districts with stronger pre-period operational capacity preserve timely payment generation and employment days per household more effectively when local workload shocks are unusually high. 5 visualizes that interaction, and the resulting benchmark floor for the local-response index is 2.0percent before the quantitative section widens that object in a bound search. This remains a reduced-form mapping from observed operational resilience into the model's local actuation object, but it is anchored in Andhra rather than treated as a free residual.

The information block remains secondary, but it is now estimated inside linked Andhra beneficiary data rather than inferred from an external scalar. Smartcard-holder households gain about 16.5percent in work access, and the linked leakage estimates are also concentrated among holders. Operationally, the benchmark chooses (η_+, η_-) so that

$$\Delta_I(\theta) = \frac{\mathcal{R}(g^*, m^*) - \mathcal{R}(g^*, 0.7m^*)}{\mathcal{R}(g^*, 0.7m^*)} \approx 7.4/100. \quad (5.7)$$

This is an Andhra-only linked-beneficiary mapping rather than an imported one. The leakage moments remain informative, but the benchmark uses the smartcard-holder access gain as the conservative structural information target.

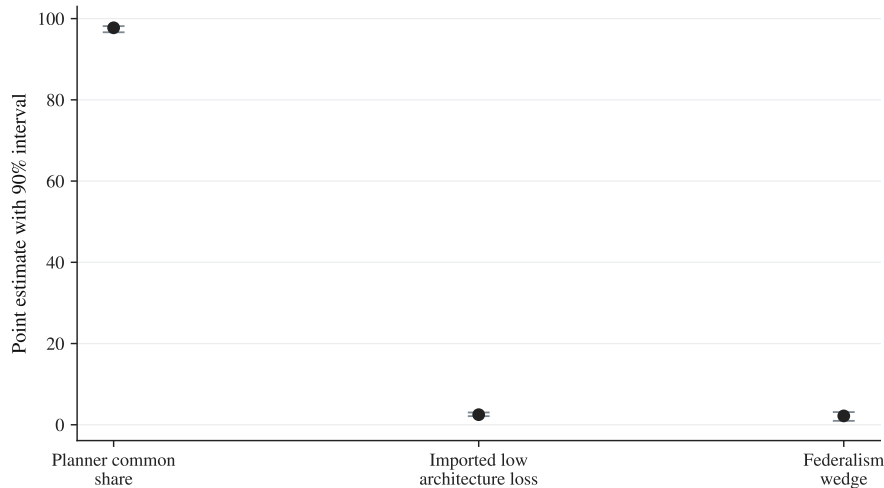


Figure 6. Main welfare conclusions survive district-bootstrap uncertainty

5.6 Structural welfare mapping

Up to this point, the empirical analysis has established two Andhra facts without the structural model: interoperable architecture improved delivery outcomes, and those gains were larger in high-common weeks. The benchmark asks what administrative portfolio rationalizes those facts in a common-shock environment. It is annual, uses log utility, and solves the steady-state system numerically. Productive capital is Cobb-Douglas, $F(k) = k^\alpha$, with output scale normalized to one. 7 reports the observed objects and fitted targets that discipline the benchmark. The key point is that the common-shock environment is anchored in observed Andhra administrative data rather than chosen for convenience.

5.7 Budget-neutral architecture comparison

The central structural counterfactual concerns composition, not the contrast between a rich administrative state and a no-capacity baseline. 8 therefore comes first. It holds total actuation and information fixed and asks what happens when a low-correlation architecture is imported from another environment. The robust conclusion is a ranking: even after propagating district-level uncertainty through the covariance and local-capacity estimates, more local architectures perform markedly worse in Andhra than common-platform architectures. The point estimate for importing the low-correlation architecture into the Andhra environment is 2.27percent of permanent consumption, and the federalism wedge remains quantitatively important across the district bootstrap.

Those losses are quantitatively important because the observed Andhra environment is far from the low-correlation case. 9 shows the corresponding benchmark architectures and scale effects. In the Andhra benchmark, the planner installs $g_C^* = 0.638$, $g_L^* = 0.015$, and $m^* = 0.358$, so the common platform accounts for about 97.7percent of actuation capacity.

7 shows the same result over the full correlation range. As needs become more common, the planner shifts decisively toward interoperable backbone capacity. The figure also reports a conservative federalism comparison in which jurisdictions internalize only one-half of the common-platform dividend while information is held fixed at the planner level. In the Andhra benchmark, that

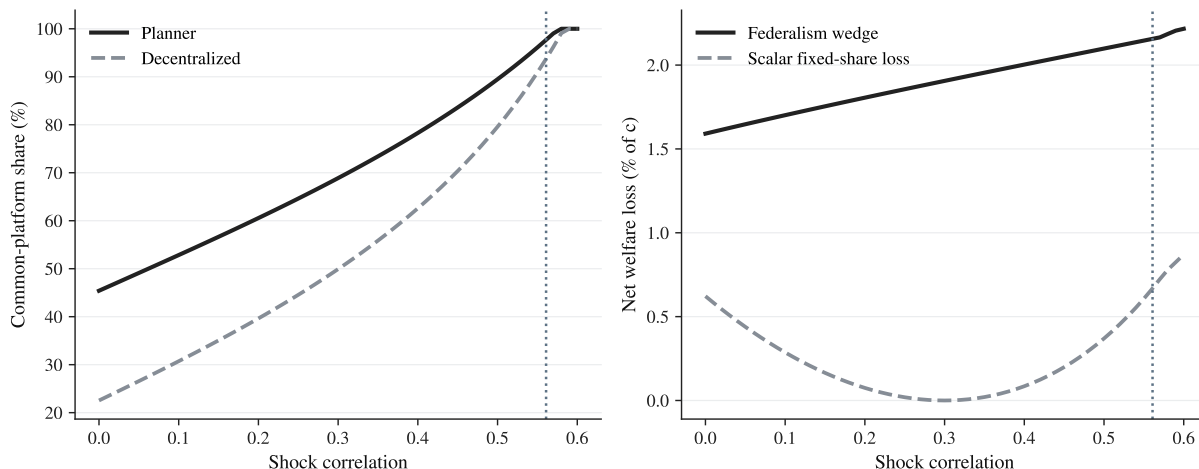


Figure 7. More common shocks raise the value of interoperable architecture

conservative wedge lowers common actuation from 0.638 to 0.327 and reduces net steady-state welfare by about 2.16 percentage points.

The information counterfactual remains secondary. Starting from the Andhra benchmark, a 10 percent increase in information capital raises gross adaptation value by the amount implied by the linked smartcard-holder information gain. The economic message is unchanged: information is more valuable when delivery capacity is stronger.

5.8 Budgeted policy counterfactual

The structural welfare numbers are easier to interpret once they are put back into observed budget language. The Andhra smartcard study reports an explicit implementation cost: a 2 percent commission paid to banks and service providers, which the study maps into about \$4.0 million in the study districts at endline scale. The same source reports about \$4.3 million in beneficiary time savings and about \$32.8 million in reduced leakage. 10 reports the corresponding budget interpretation of the architecture comparison.

6 Robustness and scope

The benchmark should be interpreted as an administrative-data policy exercise anchored first by the GP-week common-workload result and then by the raw Andhra treatment margins, SSP, and the public breadth check outside Andhra. The local and information blocks matter for the strength of the structural translation, but they are not needed to establish the existence or budget relevance of the platform effect itself. The remaining limits are therefore narrower and easier to state.

First, the local-capacity mapping still interprets observed operational resilience as a conservative reduced-form proxy for the model's local actuation object rather than as a tightly point-identified structural parameter. In Andhra, the interaction estimates are small and place the benchmark at the local-response floor of 2.0 percent. Rather than lean on that point estimate, Table 12 and

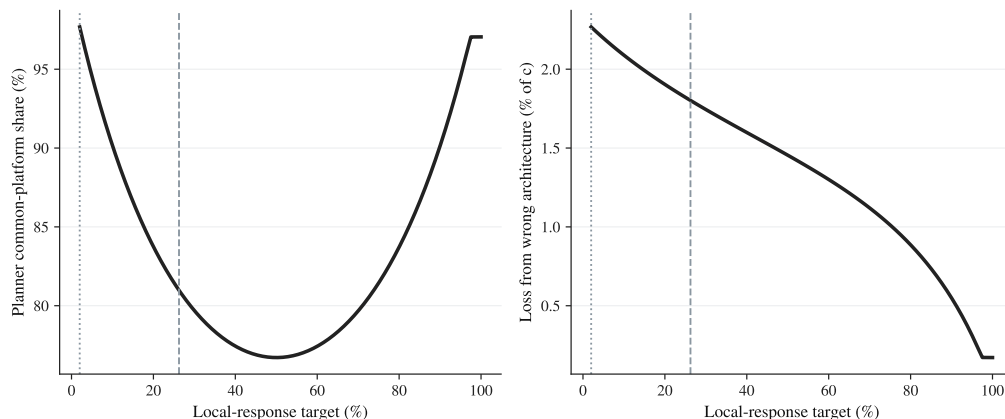


Figure 8. The Andhra common-platform ranking survives a wide local-response bound search

Figure 8 report a wide bound search that lets the local-response target range from 2.0percent up to 100.0percent, including the earlier stress point where $\hat{W}_L = \hat{W}_C$ at 26.2percent. Across that interval, the planner’s common-platform share never falls below 76.7percent and the wrong-architecture loss never falls below 0.17percent of permanent consumption. The same-setting falsifications in 11 make the remaining limit more specific: the headline interaction is not concentrated in higher-friction places, but the public files still do not deliver a clean positive-identification local-capacity design.

Second, the information block now uses linked beneficiary microdata, but it still falls short of registry-to-payment linkage that would permit a complete observability design. Third, the cross-state panel provides breadth but not a second external causal design, and the bootstrap captures district-level sampling and same-setting moment uncertainty rather than every source of policy-model misspecification. Those caveats matter, but they qualify the structural translation more than the reduced-form policy result itself.

7 Conclusion

Administrative investment is a composition problem, not a one-stock accumulation problem. In Andhra, interoperable payment and authentication architecture reduced collection time, payment delays, and payment unpredictability, while raising received payouts and work access in the raw NREGS microdata processed by this repo. The paper’s central empirical result is that the reconstructed GP-week rollout panel shows much larger gains in high-common weeks than in low-common weeks. SSP shows that the same platform improved delivery in a second program context, and the state-month panel shows a steeper readiness gradient in similarly high-common environments.

Mapping those facts into the Andhra covariance structure yields a robust ranking. Common-platform architectures dominate more local alternatives across a wide range of local-capacity assumptions, and importing the low-correlation architecture into Andhra costs about 2.3percent of permanent consumption. Even when the local-response target is allowed to rise to 100.0percent, the planner’s common-platform share never falls below 76.7percent and the wrong-architecture loss never falls below 0.17percent. The local-capacity block remains a conservative reduced-form discipline, the information block still falls short of full registry-to-payment linkage, and the repo does not yet

contain a separate external causal setting with its own covariance-based rollout design. Those limits qualify the strength of the causal decomposition, not the core architecture ranking. In common-shock environments, underinvestment in interoperable administrative backbones is likely to be costly.

A Theory map

B Public GP-week reconstruction and validation

The GP-week panel is reconstructed from the public OpenICPSR smartcards archive by combining the weekly payout file `gp-total-disbursement.dta` with the GP-level conversion file `cardedgp_nregs.dta`. The merge keys are the district, mandal, and GP identifiers `did`, `mid`, and `pid`. Weekly dates are recovered from the posted year-week fields, and conversion dates are translated from the archive’s numeric date field into calendar dates before constructing the rollout indicator. The resulting harmonized panel, `data/track_b/andhra_rollout_exposure_panel.csv`, records weekly payouts, days worked, treatment assignment, recovered conversion timing, and the post-conversion indicator for 879 GPs and 452712 GP-weeks.

The matched event-study sample requires observed pre- and post-conversion weeks for each GP. It then keeps a symmetric ± 26 week window around each recovered conversion date, yielding 560 GPs and 27010 GP-weeks in the baseline GP-week design. 1 bins event time into $[-10, -6]$, $[-5, -2]$, 0, $[1, 5]$, and $[6, 10]$ weeks and normalizes each GP by its own mean over weeks -10 through -2 so that pretrends and post-conversion divergence can be read directly from the public panel; the right-hand panel applies the same normalization to the strictest same-setting sample that keeps only original treatment-assigned GPs and defines commonality from untreated GPs outside the focal district.

Common-workload weeks are defined from leave-one-out statewide days worked, so the commonality measure does not mechanically use the focal GP’s own workload. 1 first reports the baseline decomposition on any activity, total disbursed amounts, total days worked, the overall payout-per-day outcome, and a descriptive conditional-on-positive-activity payout-per-day row, and then reports robustness to a continuous interaction, tercile and quartile ranks, alternative leave-one-out constructions based on statewide amounts and totals, an untreated-only leave-one-out statewide-days measure built only from not-yet-converted GPs in the same calendar week and dropping weeks with no untreated comparison pool, a stricter untreated-only measure built only from not-yet-converted GPs outside the focal district, a combined restriction that keeps only original treatment-assigned GPs in the outcome sample while defining common-workload from untreated GPs outside the focal district, the same combined restriction with district-specific month and year effects, a common-workload measure that excludes all GPs in the focal district from the statewide average, mandal-clustered uncertainty, leave-one-district-out leverage, a balanced-window restriction requiring at least twenty observed pre- and twenty observed post-conversion weeks per GP, exclusions of peak seasonal windows, and two placebo conversion dates.

The placebo designs assign pseudo conversion dates twelve and sixteen weeks before actual conversion and then restrict the sample to still-pre-treatment observations, so the placebo comparisons do not inherit true post-conversion weeks. The absence of comparable placebo interactions in 1 is

therefore informative about the timing reconstruction rather than about mechanical seasonality alone.

C Additional calibration note

The quantitative exercise solves a just-identified moment-matching problem for the common, local, and information blocks. Exact-fit residuals are therefore numerical diagnostics rather than independent evidence, which is why they are omitted from the main text. They are mechanically negligible in the replication output and can be recovered from the generated model summary.

D Auxiliary theoretical statements

This section records the formal results summarized in Section 3.

D.1 Actuation value and portfolio characterization

Proposition D.1 (Expected adaptation value). *Fix $g \in \mathbb{R}_+^J$ and let $W(g)$ be defined by (3.5). The quadratic recourse problem has value*

$$V(\xi, g) = -\frac{1}{2}\xi^\top (I + \kappa^{-1}W(g))^{-1}\xi. \quad (\text{D.1})$$

Relative to zero recourse, the realized gain is

$$\Delta(\xi, g) = \frac{1}{2}\xi^\top W(g)(W(g) + \kappa I)^{-1}\xi. \quad (\text{D.2})$$

If $\xi \sim \mathcal{N}(0, \Sigma)$, the ex ante adaptation value per contingency is

$$\mathcal{R}(g) = \mathbb{E} \Delta(\xi, g) = \frac{1}{2}\text{tr}[\Sigma W(g)(W(g) + \kappa I)^{-1}]. \quad (\text{D.3})$$

Proposition D.2 (Marginal values, monotonicity, and concavity). *The function $\mathcal{R}(g)$ is bounded, increasing, and concave on $\mathbb{R}_+^J$. In particular,*

$$0 \leq \mathcal{R}(g) \leq \frac{1}{2}\text{tr}(\Sigma). \quad (\text{D.4})$$

Let

$$M(g) := (W(g) + \kappa I)^{-1}.$$

Then the marginal value of actuation asset j is

$$\partial_j \mathcal{R}(g) = \frac{\kappa}{2}\text{tr}[\Sigma M(g)W_j M(g)] \geq 0. \quad (\text{D.5})$$

The inequality is strict whenever $\Sigma^{1/2}W_j \neq 0$. Moreover, if $d \in \mathbb{R}_+^J$ and $W(d) := \sum_j d_j W_j$, then the

directional derivative satisfies

$$D\mathcal{R}(g)[d] = \frac{\kappa}{2} \text{tr}[\Sigma M(g)W(d)M(g)].$$

Corollary D.3 (Mode decomposition and actuation substitution). *Suppose there exists an orthonormal basis $\{u_\ell\}_{\ell=1}^n$ such that*

$$\Sigma = \sum_{\ell=1}^n \sigma_\ell^2 u_\ell u_\ell^\top, \quad W_j = \sum_{\ell=1}^n \omega_{j\ell} u_\ell u_\ell^\top, \quad \omega_{j\ell} \geq 0. \quad (\text{D.6})$$

Define residual reachability in mode ℓ by

$$\Omega_\ell(g) := \sum_{j=1}^J g_j \omega_{j\ell}. \quad (\text{D.7})$$

Then

$$\mathcal{R}(g) = \frac{1}{2} \sum_{\ell=1}^n \sigma_\ell^2 \frac{\Omega_\ell(g)}{\kappa + \Omega_\ell(g)}, \quad (\text{D.8})$$

and the marginal value of asset j is

$$\partial_j \mathcal{R}(g) = \frac{\kappa}{2} \sum_{\ell=1}^n \frac{\sigma_\ell^2 \omega_{j\ell}}{(\kappa + \Omega_\ell(g))^2}. \quad (\text{D.9})$$

For any i, j ,

$$\partial_{ij}^2 \mathcal{R}(g) = -\kappa \sum_{\ell=1}^n \frac{\sigma_\ell^2 \omega_{i\ell} \omega_{j\ell}}{(\kappa + \Omega_\ell(g))^3} \leq 0. \quad (\text{D.10})$$

Thus actuation assets that cover the same shock directions are weak substitutes.

Corollary D.4 (When the model collapses to one-stock capacity). *Suppose there exists $\bar{W} \succeq 0$ and scalars $\omega_j \geq 0$ such that*

$$W_j = \omega_j \bar{W}, \quad j = 1, \dots, J.$$

Define the scalar effective stock $q(g) := \sum_{j=1}^J \omega_j g_j$. Then

$$W(g) = q(g) \bar{W}, \quad \mathcal{R}(g) = \tilde{\mathcal{R}}(q(g)), \quad \tilde{\mathcal{R}}(q) := \frac{1}{2} \text{tr} \left[\Sigma q \bar{W} (q \bar{W} + \kappa I)^{-1} \right].$$

Any two portfolios with the same $q(g)$ deliver the same adaptation value, and

$$\partial_j \mathcal{R}(g) = \omega_j \tilde{\mathcal{R}}'(q(g)).$$

Thus the planner's actuation problem reduces to a scalar effective-capacity choice. Architecture-specific comparative statics—including the common-versus-local tilt in 3.1—require nonproportional loadings across assets.

Proposition D.5 (Ramsey conditions with multiple capacity assets). *Along any interior optimum*

for the active actuation assets, consumption satisfies the standard Ramsey-Euler equation

$$\frac{\dot{c}_t}{c_t} = \frac{1}{\sigma} (F'(k_t) - \delta_k - \rho), \quad (\text{D.11})$$

and each active capacity asset j satisfies

$$F'(k_t) - \delta_k = -\delta_j + \frac{\lambda \partial_j \mathcal{R}(g_t)}{\nu'(c_t)}. \quad (\text{D.12})$$

Equivalently,

$$F'(k_t) - \delta_k = -\delta_j + \mathbf{m}_{jt}, \quad \mathbf{m}_{jt} := \frac{\lambda \partial_j \mathcal{R}(g_t)}{\nu'(c_t)}. \quad (\text{D.13})$$

Corollary D.6 (Steady state). *At any interior steady state (k^*, g^*, c^*) ,*

$$F'(k^*) = \rho + \delta_k, \quad (\text{D.14})$$

$$\lambda \partial_j \mathcal{R}(g^*) = (\rho + \delta_j) \nu'(c^*), \quad j = 1, \dots, J, \quad (\text{D.15})$$

$$c^* = F(k^*) - \delta_k k^* - \sum_{j=1}^J \delta_j g_j^*. \quad (\text{D.16})$$

Proposition D.7 (Shock-aligned capacity portfolio). *Under the common-eigenbasis structure in D.3, any interior steady state satisfies*

$$(\rho + \delta_j) \nu'(c^*) = \frac{\lambda \kappa}{2} \sum_{\ell=1}^n \frac{\sigma_\ell^2 \omega_{j\ell}}{(\kappa + \Omega_\ell(g^*))^2}, \quad j = 1, \dots, J. \quad (\text{D.17})$$

Hence an asset is valuable only to the extent that it covers directions with high shock variance and low residual reachability.

If, in addition, each asset is mode specific,

$$W_j = \omega_j u_j u_j^\top, \quad \delta_j \equiv \delta_g, \quad (\text{D.18})$$

then the steady-state portfolio obeys the water-filling rule

$$g_j^* = \frac{1}{\omega_j} \left[\sqrt{\frac{\lambda \kappa \sigma_j^2 \omega_j}{2(\rho + \delta_g) \nu'(c^*)}} - \kappa \right]_+, \quad j = 1, \dots, n. \quad (\text{D.19})$$

In particular, direction j is shut down when

$$\sigma_j^2 \omega_j \leq \frac{2\kappa(\rho + \delta_g) \nu'(c^*)}{\lambda}.$$

D.2 Information and decentralization

Proposition D.8 (Information dividend and complementarity). *Fix actuation portfolio g and information portfolio m . The ex ante value of recourse under posterior covariance (3.10) is*

$$\mathcal{R}(g, m) = \frac{1}{2} \text{tr} \left[(\Sigma - \Pi(m)) W(g) (W(g) + \kappa I)^{-1} \right]. \quad (\text{D.20})$$

The marginal value of information asset ℓ is

$$\partial_{m_\ell} \mathcal{R}(g, m) = \frac{1}{2} \text{tr} \left[\Pi(m) K_\ell \Pi(m) W(g) (W(g) + \kappa I)^{-1} \right] \geq 0. \quad (\text{D.21})$$

Moreover, for every actuation asset j and information asset ℓ ,

$$\partial_{g_j m_\ell}^2 \mathcal{R}(g, m) = \frac{\kappa}{2} \text{tr} [\Pi(m) K_\ell \Pi(m) M(g) W_j M(g)] \geq 0. \quad (\text{D.22})$$

Hence information and actuation are complements.

If information assets evolve according to

$$\dot{m}_{\ell t} = x_{\ell t} - \delta_\ell^m m_{\ell t}, \quad (\text{D.23})$$

and the resource constraint subtracts $\sum_{\ell=1}^L x_{\ell t}$, the planner's return condition extends immediately. Each active information asset satisfies

$$F'(k_t) - \delta_k = -\delta_\ell^m + \frac{\lambda \partial_{m_\ell} \mathcal{R}(g_t, m_t)}{\nu'(c_t)}. \quad (\text{D.24})$$

Proposition D.9 (Federalism and architecture-specific underinvestment in the matrix model). *Consider a federation of $N \geq 2$ identical jurisdictions in which each jurisdiction contributes x_i to a shared common platform $G = \sum_{i=1}^N x_i$ and chooses a local stock ℓ_i . Hold fixed any common information stock m to isolate the actuation wedge. Let the representative jurisdiction's adjustment value be*

$$\mathcal{R}_i(G, \ell_i; m) = \frac{1}{2} \text{tr} \left[(\Sigma_i - \Pi_i(m)) W_i(G, \ell_i) (W_i(G, \ell_i) + \kappa I)^{-1} \right], \quad (\text{D.25})$$

with

$$W_i(G, \ell_i) = G W_C + \ell_i W_L, \quad (\text{D.26})$$

where $W_C \succeq 0$ is the reachability object for the shared interoperable platform and $W_L \succeq 0$ is the reachability object for local execution. Under symmetry, all jurisdictions face the same (Σ_i, Π_i) and choose the same local stock ℓ .

Take the steady-state resource base per jurisdiction, net of productive-capital requirements, to be y . Then representative-jurisdiction consumption is

$$c = y - \delta_G G / N - \delta_L \ell. \quad (\text{D.27})$$

Suppose $\mathcal{R}_i(G, \ell; m)$ is increasing and concave in (G, ℓ) and the symmetric steady state is interior.

Then the social planner chooses (G, ℓ) to satisfy

$$(\rho + \delta_G)\nu'(c^{SP}) = \lambda N \partial_G \mathcal{R}_i(G^{SP}, \ell^{SP}; m), \quad (\text{D.28})$$

$$(\rho + \delta_L)\nu'(c^{SP}) = \lambda \partial_\ell \mathcal{R}_i(G^{SP}, \ell^{SP}; m). \quad (\text{D.29})$$

A symmetric decentralized equilibrium satisfies

$$(\rho + \delta_G)\nu'(c^D) = \lambda \partial_G \mathcal{R}_i(G^D, \ell^D; m), \quad (\text{D.30})$$

$$(\rho + \delta_L)\nu'(c^D) = \lambda \partial_\ell \mathcal{R}_i(G^D, \ell^D; m). \quad (\text{D.31})$$

Hence $G^D < G^{SP}$. By contrast, purely local execution capacity is not subject to the same N -fold free-riding wedge.

D.3 Welfare metric

Proposition D.10 (Exact and local welfare metrics for architecture counterfactuals). *Fix two steady-state architectures a_0 and a_1 with associated steady-state consumptions c_0 and c_1 . Let $\Delta \mathcal{R} := \mathcal{R}(a_1) - \mathcal{R}(a_0)$. The exact permanent-consumption equivalent $\tau(a_1, a_0)$ relative to architecture a_0 is defined by*

$$\nu((1 + \tau)c_0) + \lambda \mathcal{R}(a_0) = \nu(c_1) + \lambda \mathcal{R}(a_1).$$

Under CRRA utility,

$$\tau(a_1, a_0) = \begin{cases} \frac{c_1}{c_0} \exp(\lambda \Delta \mathcal{R}) - 1, & \sigma = 1, \\ \left[\left(\frac{c_1}{c_0} \right)^{1-\sigma} + (1-\sigma)c_0^{\sigma-1} \lambda \Delta \mathcal{R} \right]^{1/(1-\sigma)} - 1, & \sigma \neq 1. \end{cases} \quad (\text{D.32})$$

For nearby architectures,

$$\tau(a_1, a_0) = \frac{c_1 - c_0}{c_0} + \frac{\lambda \Delta \mathcal{R}}{c_0 \nu'(c_0)} + o(\|a_1 - a_0\|). \quad (\text{D.33})$$

For equal-budget counterfactuals with $c_1 = c_0$, only the recourse term remains.

E Proofs

Proof of D.1. Stack the fast controls as $v = (v_1, \dots, v_J)$ and define the linear operator $\mathcal{L}_g : \prod_{j=1}^J L^2([0, T]; \mathbb{R}^{m_j}) \rightarrow \mathbb{R}^n$ by

$$\mathcal{L}_g v := \int_0^T e^{A(T-s)} \sum_{j=1}^J \sqrt{g_j} B_j v_{js} \, ds.$$

Then $z_T = \mathcal{L}_g v$, and the quadratic recourse problem is equivalent to minimizing

$$J(v) = \frac{1}{2} \|\mathcal{L}_g v - \xi\|^2 + \frac{\kappa}{2} \sum_{j=1}^J \int_0^T \|v_{js}\|^2 ds.$$

The first-order condition in the product Hilbert space is

$$(\mathcal{L}_g^* \mathcal{L}_g + \kappa I) v^* = \mathcal{L}_g^* \xi.$$

Substituting the solution back into J yields

$$\min_v J(v) = \frac{1}{2} \xi^\top \left[I - \mathcal{L}_g (\mathcal{L}_g^* \mathcal{L}_g + \kappa I)^{-1} \mathcal{L}_g^* \right] \xi.$$

By the Woodbury identity,

$$\mathcal{L}_g (\mathcal{L}_g^* \mathcal{L}_g + \kappa I)^{-1} \mathcal{L}_g^* = W(g) (W(g) + \kappa I)^{-1},$$

where $W(g) = \mathcal{L}_g \mathcal{L}_g^* = \sum_j g_j W_j$ is the finite-horizon controllability Gramian. Since

$$I - W(g) (W(g) + \kappa I)^{-1} = (I + \kappa^{-1} W(g))^{-1},$$

equation (D.1) follows. The gain relative to zero recourse is (D.2). Taking expectations and using $\mathbb{E}[\xi \xi^\top] = \Sigma$ yields (3.6). $\square$

Proof of D.2. Write

$$\mathcal{R}(g) = \frac{1}{2} \text{tr}[\Sigma f(W(g))], \quad f(X) := X(X + \kappa I)^{-1} = I - \kappa(X + \kappa I)^{-1}.$$

Because $(X + \kappa I)^{-1} \preceq \kappa^{-1} I$ for $X \succeq 0$, we have $0 \preceq f(X) \preceq I$, which implies (D.4). The map $X \mapsto (X + \kappa I)^{-1}$ is operator convex and decreasing on positive semidefinite matrices, so f is operator increasing and operator concave. Since $W(g) = \sum_j g_j W_j$ is affine in g and $\Sigma \succeq 0$, the scalar function $\mathcal{R}(g)$ is increasing and concave on $\mathbb{R}_+^J$.

For the gradient, consider a direction d and differentiate the identity $f(X) = I - \kappa(X + \kappa I)^{-1}$. The Fréchet derivative is

$$Df(X)[D] = \kappa(X + \kappa I)^{-1} D (X + \kappa I)^{-1}.$$

Applying this at $X = W(g)$ and $D = W(d) := \sum_j d_j W_j$ gives

$$D\mathcal{R}(g)[d] = \frac{\kappa}{2} \text{tr}[\Sigma M(g) W(d) M(g)].$$

Setting $d = e_j$ yields (D.5). Nonnegativity follows because both $M(g) W_j M(g)$ and Σ are positive semidefinite. Strict positivity holds whenever $\Sigma^{1/2} W_j \neq 0$. $\square$

Proof of D.3. Under (D.6), the matrices commute and are diagonal in the basis $\{u_\ell\}$. Therefore

$$W(g) = \sum_{\ell=1}^n \Omega_\ell(g) u_\ell u_\ell^\top$$

with $\Omega_\ell(g)$ defined by (D.7). Plugging this decomposition into (3.6) yields

$$\mathcal{R}(g) = \frac{1}{2} \sum_{\ell=1}^n \sigma_\ell^2 \frac{\Omega_\ell(g)}{\kappa + \Omega_\ell(g)},$$

which is (D.8). Differentiating with respect to g_j gives (D.9). Differentiating once more gives (D.10). $\square$

Proof of D.4. If $W_j = \omega_j \bar{W}$ for all j , then

$$W(g) = \sum_{j=1}^J g_j W_j = \left(\sum_{j=1}^J \omega_j g_j \right) \bar{W} = q(g) \bar{W}.$$

Substituting this identity into (3.6) yields

$$\mathcal{R}(g) = \frac{1}{2} \text{tr} \left[\Sigma q(g) \bar{W} (q(g) \bar{W} + \kappa I)^{-1} \right] = \tilde{\mathcal{R}}(q(g)).$$

The gradient formula follows from the chain rule:

$$\partial_j \mathcal{R}(g) = \tilde{\mathcal{R}}'(q(g)) \partial_j q(g) = \omega_j \tilde{\mathcal{R}}'(q(g)).$$

Because $\mathcal{R}$ depends on g only through $q(g)$, any two portfolios with the same effective stock deliver the same adaptation value. $\square$

Proof of D.5. The first-order conditions for the current-value Hamiltonian are

$$\nu'(c_t) = q_t, \quad p_{jt} = q_t$$

for each active asset j . The costate equations are

$$\dot{q}_t = (\rho + \delta_k - F'(k_t)) q_t, \quad \dot{p}_{jt} = (\rho + \delta_j) p_{jt} - \lambda \partial_j \mathcal{R}(g_t).$$

Because ν is CRRA, $\dot{q}_t/q_t = -\sigma \dot{c}_t/c_t$. Combining this with the equation for $\dot{q}_t$ yields (D.11). Equating $\dot{p}_{jt}$ and $\dot{q}_t$ along an interior optimum and dividing by the common shadow value $q_t = \nu'(c_t)$ yields

$$\rho + \delta_k - F'(k_t) = \rho + \delta_j - \frac{\lambda \partial_j \mathcal{R}(g_t)}{\nu'(c_t)},$$

which rearranges to (D.12). $\square$

Proof of D.6. At an interior steady state, $\dot{c}_t = 0$, so (D.11) implies (D.14). Setting $\dot{g}_{jt} = 0$ in (3.2)

gives $i_j^* = \delta_j g_j^*$. Setting $\dot{k}_t = 0$ in (3.1) then yields (D.16). Finally, imposing (D.12) at steady state and substituting (D.14) gives (D.15). $\square$

Proof of D.7. Substituting the mode decomposition (D.9) from D.3 into the steady-state condition (D.15) yields (D.17). If each asset is mode specific as in (D.18), then $\omega_{j\ell} = 0$ for $\ell \neq j$ and $\omega_{jj} = \omega_j$. Equation (D.17) reduces to

$$(\rho + \delta_g)\nu'(c^*) = \frac{\lambda\kappa}{2} \frac{\sigma_j^2 \omega_j}{(\kappa + \omega_j g_j^*)^2}.$$

Solving for g_j^* and imposing nonnegativity yields (D.19). $\square$

Proof of 3.1. Equations (3.3) and (3.4) satisfy the common-eigenbasis structure of D.3 with two modes. Substituting those objects into (D.8) yields (3.7). Differentiating (3.7) yields the marginal values displayed in the proposition. Their monotonicity in ϱ is immediate.

At the origin,

$$\partial_{g_C} \mathcal{R}(0,0) = \frac{\sigma^2}{2\kappa} (1 + \varrho) \omega_C, \quad \partial_{g_L} \mathcal{R}(0,0) = \frac{\sigma^2}{2\kappa} (1 - \varrho) \omega_L.$$

The condition $\partial_{g_C} \mathcal{R}(0,0) > \partial_{g_L} \mathcal{R}(0,0)$ is equivalent to (3.8). If both assets are active in steady state and have the same depreciation rate, applying (D.15) to the two assets and taking ratios yields

$$\frac{(\kappa + \omega_C g_C^*)^2}{(\kappa + \omega_L g_L^*)^2} = \frac{(1 + \varrho) \omega_C}{(1 - \varrho) \omega_L},$$

which implies (3.9). $\square$

Proof of D.8. Let the posterior mean of the contingency after observing the signal be $\mu(y) := \mathbb{E}[\xi \mid y]$. Conditional on the signal, the short-run authority solves the same quadratic recourse problem as in D.1 with target $\mu(y)$ in place of ξ . Therefore the conditional gain from recourse is

$$\Delta(y; g, m) = \frac{1}{2} \mu(y)^\top W(g) (W(g) + \kappa I)^{-1} \mu(y).$$

Under linear-Gaussian updating,

$$\mathbb{E}[\mu(y) \mu(y)^\top] = \text{Var}(\mathbb{E}[\xi \mid y]) = \Sigma - \Pi(m),$$

so taking expectations yields (D.20).

For the information gradient, differentiate (3.10). Since $\Pi(m)^{-1} = \Sigma^{-1} + \sum_\ell m_\ell K_\ell$,

$$\partial_{m_\ell} \Pi(m) = -\Pi(m) K_\ell \Pi(m).$$

Differentiating (D.20) with respect to m_ℓ therefore gives (D.21). Since

$$\Pi(m) K_\ell \Pi(m) \succeq 0 \quad \text{and} \quad W(g) (W(g) + \kappa I)^{-1} \succeq 0,$$

the derivative is nonnegative.

Finally, differentiate (D.21) with respect to g_j and use the Fréchet derivative from the proof of D.2. This yields

$$\partial_{g_j m_\ell}^2 \mathcal{R}(g, m) = \frac{\kappa}{2} \text{tr}[\Pi(m) K_\ell \Pi(m) M(g) W_j M(g)],$$

which is nonnegative because all factors are positive semidefinite. $\square$

Proof of D.9. Under symmetry, the planner maximizes the representative jurisdiction's lifetime utility multiplied by N , with per-jurisdiction flow payoff

$$\nu(c) + \lambda \mathcal{R}_i(G, \ell; m), \quad c = y - \delta_G G/N - \delta_L \ell.$$

The current-value first-order conditions therefore imply

$$N \lambda \partial_G \mathcal{R}_i(G^{SP}, \ell^{SP}; m) = N \nu'(c^{SP}) \delta_G / N$$

and

$$\lambda \partial_\ell \mathcal{R}_i(G^{SP}, \ell^{SP}; m) = \nu'(c^{SP}) \delta_L,$$

which become (D.28)–(D.29) once the standard steady-state user-cost terms are reinstated.

In a symmetric decentralized equilibrium, jurisdiction i takes G_{-i} as given and chooses its own contribution x_i and local stock ℓ_i . Because $G = x_i + G_{-i}$, a unit increase in x_i raises the common platform by one unit for all jurisdictions, but the contributing jurisdiction internalizes only its own adaptation gain. The representative-jurisdiction first-order conditions are therefore

$$\lambda \partial_G \mathcal{R}_i(G^D, \ell^D; m) = \nu'(c^D) \delta_G$$

and

$$\lambda \partial_\ell \mathcal{R}_i(G^D, \ell^D; m) = \nu'(c^D) \delta_L,$$

which become (D.30)–(D.31) after reinstating the standard steady-state user-cost terms.

To compare common stocks, note that concavity of $\mathcal{R}_i$ implies that $\partial_G \mathcal{R}_i$ is decreasing in G . The planner condition multiplies that marginal adaptation value by N , whereas the decentralized condition does not. Since higher G lowers consumption and therefore raises the marginal utility cost of further common investment, the planner's first-order condition intersects at a strictly larger common stock. Hence $G^D < G^{SP}$. By contrast, the local condition has the same marginal benefit term in the planner and decentralized problems, so purely local execution capacity is not subject to the same N -fold free-riding wedge. $\square$

Proof of D.10. By definition, the exact permanent-consumption equivalent τ solves

$$\nu((1 + \tau)c_0) + \lambda \mathcal{R}(a_0) = \nu(c_1) + \lambda \mathcal{R}(a_1).$$

If $\sigma = 1$, then $\nu(c) = \log c$ and

$$\log(1 + \tau) + \log c_0 = \log c_1 + \lambda \Delta \mathcal{R},$$

so

$$1 + \tau = \frac{c_1}{c_0} \exp(\lambda \Delta \mathcal{R}),$$

which gives the first line of (D.32). If $\sigma \neq 1$, CRRA utility implies

$$\frac{((1 + \tau)c_0)^{1-\sigma} - 1}{1 - \sigma} = \frac{c_1^{1-\sigma} - 1}{1 - \sigma} + \lambda \Delta \mathcal{R}.$$

Rearranging gives

$$(1 + \tau)^{1-\sigma} c_0^{1-\sigma} = c_1^{1-\sigma} + (1 - \sigma) \lambda \Delta \mathcal{R},$$

which yields the second line of (D.32).

For the local approximation, let $\Delta c := c_1 - c_0$. A first-order Taylor expansion around c_0 gives

$$\nu(c_1) - \nu(c_0) = \nu'(c_0) \Delta c + o(\|a_1 - a_0\|).$$

By definition of τ ,

$$\nu((1 + \tau)c_0) - \nu(c_0) = \nu(c_1) - \nu(c_0) + \lambda \Delta \mathcal{R}.$$

Expanding the left-hand side around c_0 yields

$$\nu'(c_0) \tau c_0 = \nu'(c_0) \Delta c + \lambda \Delta \mathcal{R} + o(\|a_1 - a_0\|).$$

Dividing by $c_0 \nu'(c_0)$ gives (D.33). If $c_1 = c_0$, the consumption term disappears. $\square$

F Benchmark details

For the two-region benchmark in Section 4, let

$$\sigma_+^2 = s_\xi^2(1 + \varrho), \quad \sigma_-^2 = s_\xi^2(1 - \varrho), \quad (\text{F.1})$$

$$\pi_+(m) = (\sigma_+^{-2} + \eta_+ m)^{-1}, \quad \pi_-(m) = (\sigma_-^{-2} + \eta_- m)^{-1}. \quad (\text{F.2})$$

The adaptation value is

$$\mathcal{R}(g_C, g_L, m; \varrho) = \frac{1}{2} \left[(\sigma_+^2 - \pi_+(m)) \frac{\omega_C g_C}{\kappa + \omega_C g_C} + (\sigma_-^2 - \pi_-(m)) \frac{\omega_L g_L}{\kappa + \omega_L g_L} \right]. \quad (\text{F.3})$$

Under log utility and Cobb-Douglas production with scale normalized to one, the steady-state productive capital stock satisfies

$$k^* = \left(\frac{\alpha}{\rho + \delta_k} \right)^{1/(1-\alpha)}. \quad (\text{F.4})$$

The planner's numerical system is

$$c = (k^*)^\alpha - \delta_k k^* - \delta_C g_C - \delta_L g_L - \delta_m m, \quad (\text{F.5})$$

$$\frac{\rho + \delta_C}{c} = \frac{\lambda \kappa}{2} (\sigma_+^2 - \pi_+(m)) \frac{\omega_C}{(\kappa + \omega_C g_C)^2}, \quad (\text{F.6})$$

$$\frac{\rho + \delta_L}{c} = \frac{\lambda \kappa}{2} (\sigma_-^2 - \pi_-(m)) \frac{\omega_L}{(\kappa + \omega_L g_L)^2}, \quad (\text{F.7})$$

$$\frac{\rho + \delta_m}{c} = \frac{\lambda}{2} \left[\pi_+(m)^2 \eta_+ \frac{\omega_C g_C}{\kappa + \omega_C g_C} + \pi_-(m)^2 \eta_- \frac{\omega_L g_L}{\kappa + \omega_L g_L} \right]. \quad (\text{F.8})$$

The conservative decentralized benchmark in Section 4 fixes information at the planner value m^* and solves

$$c^D = (k^*)^\alpha - \delta_k k^* - \delta_C g_C^D - \delta_L g_L^D - \delta_m m^*, \quad (\text{F.9})$$

$$\frac{\rho + \delta_C}{c^D} = \frac{\lambda}{N} \partial_{g_C} \mathcal{R}(g_C^D, g_L^D, m^*; \varrho), \quad N = 2, \quad (\text{F.10})$$

$$\frac{\rho + \delta_L}{c^D} = \lambda \partial_{g_L} \mathcal{R}(g_C^D, g_L^D, m^*; \varrho). \quad (\text{F.11})$$

All figures are generated by solving these systems directly and tracing the solution over a grid of correlation values ϱ .

G Task-counting microfoundation for reachability volume

An alternative log-determinant index of the controllability Gramian remains available as a special case of the present framework.

Proposition G.1 (Task counting and volume of reachable states). *Fix an energy budget $\bar{e} > 0$. Suppose society receives one unit of payoff when a terminal target can be reached exactly within control energy at most $\bar{e}$, and zero otherwise. For the frozen fast system, the set of targets achievable under that budget is*

$$\mathcal{E}_T(\bar{e}; g) = \{\xi \in \mathbb{R}^n : \xi^\top W(g)^{-1} \xi \leq \bar{e}\}.$$

If tasks are uniformly distributed over a neighborhood containing $\mathcal{E}_T(\bar{e}; g)$, then the mass of solvable tasks is proportional to

$$\text{vol}(\mathcal{E}_T(\bar{e}; g)) = \kappa_n \bar{e}^{n/2} \sqrt{\det W(g)},$$

where κ_n is the volume of the unit ball in $\mathbb{R}^n$. Hence $\frac{1}{2} \log \det W(g)$ is an exact monotone index of value in this special case.

Proof. For the frozen linear system with initial condition $z_0 = 0$, the minimum control energy needed to reach target ξ at horizon T is $\xi^\top W(g)^{-1} \xi$. Hence the set of reachable targets within energy budget $\bar{e}$ is the ellipsoid $\mathcal{E}_T(\bar{e}; g)$. The volume of an ellipsoid $\{x : x^\top A^{-1} x \leq \bar{e}\}$ is $\kappa_n \bar{e}^{n/2} \sqrt{\det A}$, which gives the formula above with $A = W(g)$. $\square$

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Table 1. Headline mechanism test: GP-week rollout gains are larger in high-common weeks, chiefly by reducing zero-activity weeks and raising total disbursements

Outcome / specification	Low-common rollout effect	High-common rollout effect	High-minus-low differential
<i>Panel A. Baseline decomposition in the matched Andhra GP-week panel</i>			
Any disbursement / work this week (pp)	5.6 (1.7)	15.1 (1.6)	9.6 (1.8)
Log weekly amount disbursed	0.439 (0.170)	1.531 (0.169)	1.092 (0.184)
Log weekly days worked	0.187 (0.095)	0.878 (0.100)	0.691 (0.105)
Log payout per day worked (overall GP-week outcome)	0.254 (0.080)	0.657 (0.074)	0.403 (0.085)
Log payout per day worked, conditional on positive-work weeks	-0.010 (0.013)	-0.048 (0.011)	-0.039 (0.012)
<i>Panel B. Robustness of the headline payout-per-day differential</i>			
Continuous common-workload interaction (one s.d.)	—	—	0.153 (0.034)
Tercile-rank interaction (top vs. bottom)	—	—	0.139 (0.054)
Quartile-rank interaction (top vs. bottom)	—	—	0.100 (0.038)
Alternative leave-one-out amount measure	—	—	0.494 (0.089)
Alternative leave-one-out total-days measure	—	—	0.396 (0.087)
Restrict to original treatment-assigned GPs	—	—	0.450 (0.088)
Untreated-only leave-one-out days measure	—	—	0.358 (0.086)
Untreated-only measure excluding the focal district	—	—	0.519 (0.095)
Treatment-assigned sample + untreated outside-district measure	—	—	0.596 (0.097)
Strictest sample + district month/year effects	—	—	0.409 (0.095)
Exclude focal district from common-workload measure	—	—	0.509 (0.103)
Mandal-clustered standard errors	—	—	0.403 (0.110)
Leave-one-district-out minimum across districts	—	—	0.250 (0.090)
Exclude April–June weeks	—	—	0.311 (0.102)
Exclude July–September weeks	—	—	0.198 (0.093)
Require at least 20 observed pre and 20 observed post weeks	—	—	0.426 (0.092)
Placebo conversion date (12 weeks early)	—	—	-0.055 (0.128)
Placebo conversion date (16 weeks early)	—	—	0.032 (0.127)

Notes: Standard errors are in parentheses and clustered by GP unless otherwise stated. Panel A uses the matched ± 26 week Andhra GP-week panel with GP fixed effects and calendar month and year effects, covering 560 GPs in 8 districts and 53 mandals. The first row is a linear-probability outcome reported in percentage points. High-common weeks are above the matched-sample median of the leave-one-out statewide log average-days measure, so the split does not use the focal GP's own workload. The conditional payout-per-day row is descriptive and shows that the overall amplification mainly reflects fewer zero-activity weeks and larger total disbursements rather than a clean within-active-week payout-rate gain. Panel B keeps the headline overall log payout-per-day outcome and adds treated-assignment-only, untreated-only, untreated-only-outside-district, the combined treated-assignment plus untreated-outside-district restriction, the same combined restriction with district-specific month and year effects, outside-district, mandal-clustered, leave-one-district-out, balanced-window, season-window, and placebo checks. The treated-assignment-only row restricts the sample to the 496 GPs originally assigned to treatment, and the combined rows keep that outcome sample while defining common-workload only from untreated GPs outside the focal district. Leave-one-district-out interactions range from 0.250 to 0.480 across the 8 Andhra districts in the matched panel.

Table 2. Repo-estimated NREGS causal effects from the raw Andhra microdata

Observed policy margin	Andhra estimate	Policy reading
Collection time per payment	19.4% lower	Interoperable payout access materially lowers the time needed to collect an NREGS payment.
Payment delay after work	25.2% lower	The common platform speeds the conversion of approved work into received payments.
Payment unpredictability	34.1% lower	Delivery timing becomes more reliable, not only faster on average.
Survey-reported payout received	23.9% higher	Beneficiaries report receiving larger NREGS payouts, consistent with lower slippage on the treated margin.
Work access	16.8% higher	Household participation rises on a first-order policy outcome rather than only on process margins.
Beneficiary time savings relative to observed platform cost	1.07 $\times$ cost	Time savings alone roughly repay the observed implementation budget.

Table 3. Supporting breadth check: SSP shows delivery gains in a second program on the same platform

Observed SSP margin	Estimate	Reading
Carded payment share	59.4 pp	Official first stage from the raw SSP GP file confirms large platform take-up in the pension program.
Collection time per payment	4.5% lower	The same smartcard backbone also lowers collection time in SSP, even though the magnitude is smaller than in NREGS.
Survey-reported amount received	5.0% higher	Pension recipients report higher amounts received under the platform.
Leakage	51.5% lower	SSP confirms that the platform improves delivery in a different beneficiary and payment environment.

Table 4. Supporting breadth check outside Andhra (correlational): platform readiness matters more where shocks are more common

Statistic	Estimate	Interpretation
Platform readiness proxy	-10.050 (3.376)	Association between readiness and the public payment-timeliness metric at zero common-shock exposure.
Common-shock share	-11.976 (4.313)	Level association between more common workload fluctuations and the public payment-timeliness metric.
Readiness $\times$ common-shock share	16.989 (5.682)	Positive interaction means readiness is more strongly associated with payment performance in high-common states.
Implied readiness slope at the 25th percentile of common-shock exposure	-3.222	Low-common states show a shallower readiness gradient.
Implied readiness slope at the 75th percentile of common-shock exposure	0.426	High-common states show a steeper readiness gradient.
State-month observations	2,032	State-month regression sample size.
States in regression sample	32	States with non-missing common-shock and readiness measures.

Notes: This table is a correlational breadth check rather than a second causal design. The dependent variable is the winsorized log of the public MGNREGA payment-timeliness measure. The regression includes calendar-month and fiscal-year effects, with standard errors clustered by state.

Table 5. Authenticated-channel use rises sharply for holders, and NREGS leakage gains are concentrated there

Outcome	Holders	Non-holders	Holder minus non-holder	Reading
Recent smartcard payment, NREGS (pp)	78.5 (3.5)	25.8 (5.0)	52.7 (4.7)	Treatment moves observed authenticated payment use much more for smartcard holders than for non-holders.
Recent smartcard payment, SSP (pp)	76.7 (7.0)	35.2 (8.4)	41.5 (5.1)	The same authenticated-channel first stage appears in the pension program as well.
NREGS work access (pp)	10.3 (4.9)	5.7 (6.1)	4.5 (5.0)	Access gains remain positive for both groups, so work access alone is not the sharpest authentication discriminator.
NREGS leakage reduction	Rs. 64.4 (29.9) lower	Rs. 26.0 (22.0) higher	Rs. 90.4 (26.0) lower	Anti-leakage gains are concentrated where authenticated smartcard use rises most.

Table 6. Policy-relevant Andhra empirical objects used in the benchmark

Object	Same-setting evidence	Estimand used in the model	Status
Shock covariance $\hat{\Sigma}$	Andhra district-month MGNREGA personday flows from the public data.gov.in feed, residualized for district, month, and fiscal-year effects	Leading common-mode share of the full district covariance matrix implies $\hat{\rho} = 0.561$ and a common component explaining 57.8% of residual variance	Same-setting district-month estimate
Common-platform response $\hat{W}_C$	Repo-estimated NREGS treatment effects on collection time, payment delay, and payment predictability from the raw OpenICPSR survey files	Mean payment-process response of 26.2% at an observed carded-payment share of 44.7%	Same-setting raw-microdata estimate
Local response $\hat{W}_L$	Andhra district interaction design: idiosyncratic monthly workload shocks interacted with pre-period district operational capacity	Average local-response index of 2.0%, combining the interaction effects on payment timeliness and days of employment per household	Same-setting interaction discipline; benchmark floor plus wide bound search, not a clean positive-ID local design
Information/authentication gain $\hat{K}$	Linked NREGS and SSP beneficiary microdata: treatment sharply raises observed smartcard-payment use for holders relative to non-holders, and NREGS leakage reductions are concentrated among holders	Effective information gain of 7.4% of adaptation value in the benchmark mapping, still anchored on the smartcard-holder work-access gain	Same-setting usage-and-leakage bridge; still not direct registry-to-payment linkage

Table 7. Observed objects and calibrated technology targets in the Andhra benchmark

Quantity	Value	Role in the benchmark
Estimated shock correlation $\hat{\rho}$	0.561	Pins down the common-versus-idiosyncratic composition of adjustment needs from Andhra district-month data.
Common-platform response $\hat{W}_C$	26.2%	Target for the fitted common reachability parameter ω_C .
Local-capacity response $\hat{W}_L$	2.0%	Target for the fitted local reachability parameter ω_L .
Information/authentication gain $\hat{K}$	7.4%	Target for the fitted information precision parameters (η_+, η_-) .
Calibrated planner common share	97.7%	Implied benchmark architecture in the Andhra environment.

Table 8. Structural welfare mapping of the Andhra policy choice

Quantity	Point estimate	90% district-bootstrap interval
Planner common-platform share in Andhra (% of actuation)	97.7	[96.6, 98.1]
Equal-budget loss from importing the low-correlation architecture (% of consumption)	2.27	[2.12, 3.03]
Federalism wedge in the Andhra benchmark (percentage points of net welfare)	2.16	[0.97, 3.12]

Table 9. Budget-neutral architecture comparison across empirical environments

Scenario	g_C	g_L	m	Common share (%)	Net CEV (% of c)
Low-correlation planner	0.364	0.437	0.308	45.4	6.43
Andhra planner	0.638	0.015	0.358	97.7	9.05
Decentralized Andhra benchmark	0.327	0.022	0.358	93.7	6.89

Table 10. Budgeted policy interpretation of the Andhra architecture choice

Policy comparison	Budget anchor	Quantitative implication
Observed interoperable payment-and-authentication platform in the study districts	\$ 4.0 million	Annualized endline-scale platform cost implied by the 2% commission paid to implementing banks and service providers in the Andhra smartcard study.
Beneficiary time savings from the platform	\$ 4.3 million	Time savings alone are about 1.07 times observed platform cost, implying near break-even efficiency even before leakage gains.
Leakage reduction on the carded margin	\$ 32.8 million	Observed anti-leakage gains are about 8.20 times platform cost, indicating substantial fiscal returns to the common architecture in Andhra.
Budget-neutral architecture mistake in Andhra	Fixed actuation and information envelope	Importing the low-correlation, more local architecture into Andhra loses 2.27% of permanent consumption.
Conservative underbuilt-backbone benchmark	Same information stock, weaker common platform	The federalism wedge lowers net welfare by 2.16 percentage points in the Andhra benchmark.

Table 11. Same-setting falsifications against local-frictions or rollout-timing-selection readings of the Andhra GP-week result

Statistic	Estimate	Interpretation
<i>Panel A. GP-week common-shock amplification by pre-treatment operational-friction proxy</i>		
Low baseline collection-time GPs	0.374 (0.119)	The headline GP-week interaction is already large where baseline collection frictions were lower.
High baseline collection-time GPs	0.431 (0.121)	Worse pre-treatment collection frictions do not generate a distinctly larger common-shock interaction.
Triple difference: high versus low collection-time GPs	0.081 (0.160)	If the Andhra result were mainly a weak-local-capacity story, this differential should be much larger and more precise than it is.
Low baseline delay-dispersion mandals	0.592 (0.130)	The GP-week amplification is also present in places with lower pre-period delay dispersion.
High baseline delay-dispersion mandals	0.380 (0.132)	Higher pre-period payout dispersion does not create a stronger same-setting interaction.
Triple difference: high versus low delay-dispersion mandals	0.136 (0.179)	This second pre-treatment proxy again fails to show that the Andhra GP-week result is concentrated in operationally weaker places.
<i>Panel B. Recovered rollout timing within the original treatment-assigned GP sample</i>		
Baseline collection time (weeks later per 1 s.d.)	-0.0 (1.0)	Slow-collection GPs do not convert meaningfully later once district fixed effects are absorbed.
Baseline survey payout amount (weeks later per 1 s.d.)	1.2 (1.5)	Recovered conversion dates are not systematically shifted toward initially higher-payout GPs.
Baseline work access (weeks later per 1 s.d.)	0.6 (1.0)	Recovered timing is not meaningfully sorted on baseline work access either.

Notes: Panel A entries are coefficients on the high-common rollout interaction in the matched ± 26 week Andhra GP-week panel. The baseline collection-time proxy covers 560 of 560 matched GPs, while the delay-dispersion proxy covers 461 of 560 matched GPs. The triple-difference rows compare the high-common rollout interaction in higher-friction versus lower-friction groups. Panel B entries regress recovered conversion timing on one standardized pre-treatment GP covariate at a time inside the 496 original treatment-assigned GPs in the matched Andhra panel, with district fixed effects and mandal-clustered standard errors; the earliest-to-latest recovered conversion span in that sample is 133 weeks.

Table 12. The common-platform ranking survives a wide local-capacity bound search

	Andhra benchmark floor	Stress point: set $\hat{W}_L = \hat{W}_C$	Wide search over local-response targets
Local-response target $\hat{W}_L$	2.0%	26.2%	Search interval up to 100.0%
Planner common-platform share	97.7%	81.0%	Never below 76.7%
Equal-budget loss from importing the low-correlation architecture	2.27%	1.80%	Never below 0.17%
Break-even local target in searched interval	—	—	None found

Notes: The Andhra local interaction estimates are economically small and place the benchmark at the conservative floor of 2.0 percent. The middle column raises the local-response target to the Andhra common-platform response. The right-hand column reports minima over the full 2.0 to 100.0 percent search interval; the minimum common-platform share and minimum architecture loss occur at different local-response targets, so the bound column is intentionally conservative.

Table 13. Appendix theory map

Core result	Policy content	Empirical role
Common-shock comparative static	More correlated needs raise the value of interoperable common capacity relative to purely local redundancy.	Motivates the Andhra district-month covariance estimate and the correlation counterfactual.
Information-actuation complementarity	Authentication, registries, and data systems are more valuable when delivery can act on what they reveal.	Maps the smartcard leakage and authentication evidence into the information block.
Federalism wedge	Shared backbones are underprovided when jurisdictions internalize only a fraction of the common dividend.	Motivates the decentralized comparison and the reported federalism wedge.
Welfare metric	Converts observed delivery margins into budget-neutral architecture losses in the structural mapping.	Produces the Andhra structural welfare comparison reported in the main text.

Table 14. Public data layers behind the GP-week and breadth exercises

Public data layer	Quantitative scope	Policy role
Repo-estimated NREGS and SSP effect file	21 program-outcome estimates	Produces the paper’s headline causal NREGS evidence and the SSP second-context estimates directly from the raw OpenICPSR survey and leakage files.
Linked beneficiary information panel	21,504 beneficiary and household observations	Supports direct smartcard-holder versus non-holder evidence on access and leakage.
Andhra GP-week payout panel	452,712 GP-weeks across 879 GPs	Direct public rollout-exposure panel from OpenICPSR with weekly payouts, days worked, treatment assignment, and recovered conversion timing.
Matched Andhra conversion panel	87,360 GP-weeks across 560 GPs	Supports within-GP rollout-timing analyses using observed weeks around recovered conversion dates.
Cross-state validation panel	2,160 state-months across 32 states	Places Andhra in a broader administrative environment using public MGNREGA common-shock statistics and ABPS readiness proxies.
Andhra’s position in the cross-state distribution	Common-shock rank 11/32; readiness rank 1/32	Andhra lies in the high-readiness, above-median common-shock portion of the public cross-state distribution.